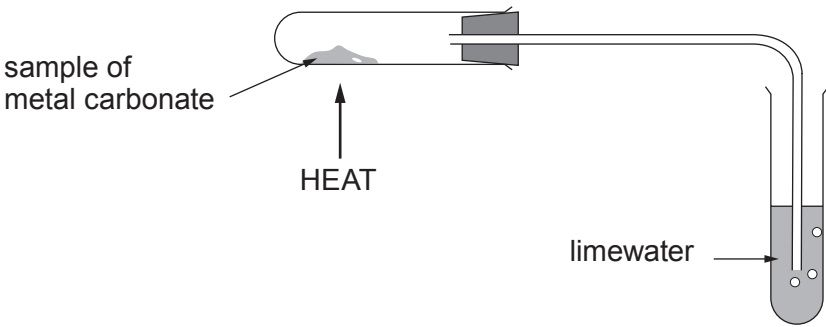


6. (a) A student investigated the decomposition of three different metal carbonates.
- She measured the time taken for limewater to turn milky using the following apparatus.



Three samples of each metal carbonate were tested. Her results are shown in the table.

Metal carbonate	Time taken for limewater to turn milky (s)			
	Sample 1	Sample 2	Sample 3	Mean
copper(II) carbonate	15	25	17	
zinc carbonate	54	52	53	53
calcium carbonate	195	200	190	195

- (i) Calculate the mean time taken for limewater to turn milky on heating copper(II) carbonate. **Show your working.** [2]

Mean time = s



- (ii) I. Place the carbonates in order of stability giving a reason for your answer. [2]

Most stable

.....

Least stable

Reason

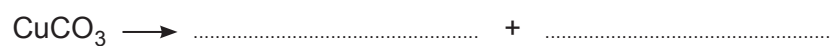
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- II. Explain the order of stability of the carbonates. [1]

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- (iii) Complete the following symbol equation for the decomposition of copper(II) carbonate. [2]



- (b) Calculate the relative formula mass, M_r , of copper(II) carbonate, CuCO_3 . [2]

$$A_r(\text{C}) = 12 \quad A_r(\text{O}) = 16 \quad A_r(\text{Cu}) = 63.5$$

$$M_r = \dots$$

